

Technology Stack

Hospital Management System

A complete reference of every tool, language, framework, and service used to build MedCore HMS — a modular, ABDM-ready hospital management system for small & mid-size Indian clinics and hospitals. Every component is free and open-source.

Project: MedCore HMS • Modular monolith (feature-flag based)

Team: 2 developers • **Deployment:** Single-tenant, on-premise

Repository: hms-software • **Working folder:** D:\HMS software

Document version: 1.0 • **Prepared for:** Suhas

1. Stack at a Glance

This project uses a deliberately simple, proven, and 100% free toolset. Nothing here requires a paid licence. The only real-world costs are a domain name, optional off-site backup storage, and per-transaction fees if you later add SMS or an online payment gateway. The stack is organised into six layers, described in the following sections.

Layer	Main technologies	What it does (plain words)
Backend	Python, FastAPI, SQLAlchemy, Alembic	The engine — does the work behind the scenes
Database	PostgreSQL, Valkey	Where all information is stored and cached
Frontend	React, Vite, TypeScript, Ant Design	The screens staff see and click
Security & Auth	Own JWT login, Argon2, Caddy (TLS)	Logins, roles, encryption
Infrastructure	Docker, Docker Compose, Ubuntu Server	Runs everything reliably on one server
Dev & Testing	Git, GitHub, pytest, VS Code + Claude Code	Writing, testing, and version-tracking the code

Core / mandatory

Optional / later phase

Client-side

2. Backend — The Engine

This is the part users never see. It receives requests (e.g. "save this patient"), applies the rules, talks to the database, and sends answers back. Written in Python because it is beginner-friendly and has the richest ecosystem for healthcare and data work.

Tool	Version	Cost	Purpose
Python	3.12+	Free	The main backend programming language
FastAPI	0.11x	Free	Framework that builds the API (the menu of actions the frontend can request)
Uvicorn	0.3x	Free	The server program that actually runs FastAPI
SQLAlchemy	2.0.x	Free	Lets Python talk to the database without writing raw SQL by hand
Alembic	1.14+	Free	Tracks and applies changes to the database structure over time
Pydantic	2.x	Free	Checks that incoming data is valid (e.g. an age is a number)
psycopg	3.2.x	Free	The driver that connects Python to PostgreSQL
Celery	5.4+	Free	Runs slow background jobs (alerts, reports) without freezing the app
python-multipart	latest	Free	Handles file uploads (e.g. lab report PDFs)

3. Database — Where Information Lives

Every patient, bill, prescription, and stock item is stored here. PostgreSQL is the storage vault; Valkey is a fast "scratchpad" used for temporary data and speeding things up.

Tool	Version	Cost	Purpose
PostgreSQL	16 / 18	Free	The main database — reliable, powerful, industry standard for healthcare data
Valkey	8.x	Free	In-memory cache and job queue (an open, free fork of Redis)

Note: Your live decision was PostgreSQL 16 for stability during setup; the roadmap allows moving to 18 later. Either works.

4. Frontend — The Screens Users See

This is everything visible: the login page, patient registration form, billing screen, dashboards. Built with React, which is the most widely used tool for modern web interfaces and the best supported by AI coding assistants.

Tool	Version	Cost	Purpose
React	19.x	Free	The core library for building interactive screens
Vite	7.x	Free	The build tool & dev server — shows your changes instantly while coding
TypeScript	5.x	Free	A safer version of JavaScript that catches mistakes early
Ant Design	5.x	Free	Ready-made buttons, tables, forms — so you don't design from scratch
TanStack Query	v5	Free	Fetches data from the backend and keeps screens up to date
React Router	v7	Free	Handles moving between screens (pages)
Recharts	latest	Free	Draws charts and graphs on dashboards
dayjs	latest	Free	Handles dates and times cleanly
axios	latest	Free	Sends requests from the screens to the backend
i18next	latest	Free	Regional-language support (later phase)

Later-phase client apps

Tool	Version	Cost	Purpose
Expo / React Native	SDK 53+	Free	Patient mobile app for Android/iOS later
Tauri	2.x	Free	Wraps the web app into an installable desktop program later

5. Security, Authentication & Compliance

Healthcare data is sensitive and legally protected under India's DPDP Act. These tools handle who can log in, what each role is allowed to see, and keep data encrypted.

Tool	Version	Cost	Purpose
Own JWT login	—	Free	Custom-built username/password login with roles (chosen over Keycloak for simplicity)
python-jose	latest	Free	Creates the secure login "tokens" that prove who you are
Argon2 (argon2-cffi)	latest	Free	Scrambles passwords so they can never be read, even by you
Caddy	2.x	Free	Handles HTTPS automatically — encrypts data travelling over the network
pgcrypto	built-in	Free	Encrypts sensitive columns inside the database
Audit log	custom	Free	A tamper-proof record of who viewed or changed each patient record

Compliance targets built in from day one: DPDP Act (data protection), ABDM/FHIR readiness (national health standards), GST-correct billing.

6. Infrastructure — Running It All

Docker packs the whole system into standard "containers" so it runs identically on your laptop and on every hospital's server. This is what lets a two-person team support many installations without chaos.

Tool	Version	Cost	Purpose
Docker Engine	latest	Free	Runs each part of the system in its own isolated "container"
Docker Compose	v2	Free	Starts the whole stack (database + app + cache) with one command
Ubuntu Server	24.04 LTS	Free	The operating system on each hospital's server
Caddy	2.x	Free	Reverse proxy + automatic HTTPS certificates
rclone	latest	Free	Copies encrypted backups to off-site storage

7. Development, Testing & Monitoring

The tools you and your teammate use every day to write, test, track, and watch over the software.

Tool	Version	Cost	Purpose
VS Code	latest	Free	The editor where you write all the code
Claude Code	extension	Free*	AI assistant inside VS Code that writes and fixes code for you
Git	2.4x	Free	Tracks every change; unlimited undo history
GitHub	—	Free	Online backup and shared workspace for the two of you
GitHub Actions	—	Free tier	Automatically checks the code every time you save to GitHub
pytest + httpx	latest	Free	Automated tests for the backend
Playwright	latest	Free	Automated tests that click through the screens like a real user
Locust	latest	Free	Tests whether the system stays fast under many users
Prometheus + Grafana	3.x / 12.x	Free	Monitors health of every hospital install later
Uptime Kuma	1.23+	Free	Alerts you if a hospital's system goes down later
GlitchTip	latest	Free	Records errors so you can fix them later

* Claude Code usage may depend on your Anthropic plan; the VS Code extension itself is free to install.

8. Documents, Reports & Payments

Tool	Version	Cost	Purpose
WeasyPrint / ReportLab	latest	Free	Generates printable PDF bills, receipts, prescriptions (A4 + thermal)
openpyxl / pandas	latest	Free	Creates Excel reports and exports
fhir.resources	latest	Free	Builds ABDM-standard health records (FHIR) later
HAPI FHIR	latest	Free	Validates those health records later
Razorpay / UPI SDK	latest	Free SDK**	Online payments optional

** The payment SDK is free; payment gateways charge a small per-transaction fee. Cash and UPI-QR collection have no software cost.

9. Cost Summary

Total software cost to build this project: ₹0.

Every language, framework, database, and tool listed is free and open-source. The only unavoidable real-world costs are:

- A domain name (~₹800–1,500 / year, only when you go live)
- Optional off-site backup storage (a few ₹ per hospital, or use the hospital's own second disk at no cost)
- Optional per-transaction fees for SMS or online card/UPI payment gateways (only if you enable them)

MedCore HMS — Technology Stack Reference • Version 1.0 • Prepared for Suhas. Tool versions reflect current stable releases as of mid-2026; pin exact versions at install time and re-check official sources, as patch releases are frequent. This document lists tooling only and is not legal, financial, or compliance advice.